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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI Assisted Data Visualization for Business Decision Support Systems	
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Abstract

The rapid growth of business data has created a need for intelligent systems capable of transforming raw information into meaningful insights for effective decision-making. This project, "AI-Assisted Data Visualization for Business Decision Support Systems," aims to develop an intelligent analytics platform that combines data science, machine learning, and interactive visualization techniques to support business decision-making processes. The project utilizes benchmark business datasets, including the Superstore Sales Dataset, Global Superstore Dataset, and Brazilian E-Commerce Public Dataset (Olist), to analyze sales performance, customer behavior, profitability, and regional business trends.

The proposed system follows a systematic data science pipeline consisting of data collection, preprocessing, exploratory data analysis (EDA), feature engineering, feature selection, machine learning model development, explainable artificial intelligence (XAI), and interactive dashboard visualization. Advanced machine learning algorithms such as Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost are employed to generate predictive business insights, while SHAP and LIME techniques provide transparent explanations for model predictions. An interactive dashboard developed using Streamlit enables users to visualize key performance indicators (KPIs), monitor business trends, and explore analytical results in real time.

The primary objective of this project is to bridge the gap between complex data analytics and business decision-making by providing an AI-assisted visualization platform that delivers accurate predictions, meaningful insights, and actionable recommendations. The proposed system is expected to enhance organizational decision-making, improve business performance analysis, and demonstrate the practical application of artificial intelligence and data visualization techniques in modern business intelligence systems.

Signature of the Guide HOD